SECTION 208

DRAINS FOR ROCK CUTS

DESCRIPTION

208.01 Scope - This Section covers the installation of lined drains in the designated areas to the specified depth. The exact locations of drain holes will be dependent on field conditions and will be detailed by the Designer.

MATERIALS

208.11 PVC Pipe Materials - Drain hole lining shall consist of Schedule 40 slotted PVC pipe. The outside diameter of the pipe shall not be less than 67% of the borehole diameter, and the internal diameter of the pipe shall be a minimum of 19 mm. The slots shall be along the entire length of the pipe in one row. Unless otherwise specified, the slots shall be 0.5 mm wide, spaced 6 mm apart and have a minimum length of 30% of the outside circumference of the pipe.

CONSTRUCTION

208.31 Drilling - Advancement of drilling for drain holes shall be in an upslope direction. The dip angle for drain holes shall be 5°, or as specified by the Designer. Finished holes shall be flushed with air to remove deleterious materials.

208.32 PVC Pipe Installation: PVC pipes shall be installed into the finished holes immediately after flushing. Pipes shall be coupled according to the manufacturer's specifications. The pipe shall be installed along the entire length of the holes and shall protrude 0.30 m out of the rock face. The slots shall be on top when the pipe is installed. The pipe shall be secured in the hole, using plastic or wooden wedges driven into the hole collar if necessary, so that the pipe cannot be pulled out by hand.

SECTION 209

SHOTCRETE FOR ROCK CUTS

Note: For any SS Drawing (SP XXX - XX) referenced in this Section refer to the like-numbered Drawings in the "Standard Specifications For Highway Construction" 2006 edition.

DESCRIPTION

209.01 General - Unless otherwise specified, the shotcrete standards of ACI 506.2 and the standards of good practice in ACI 506R shall apply. DBSS 211, Portland Cement Concrete, shall also apply for shotcrete, unless otherwise specified.

Shotcrete operations shall include, but not be limited to pre-construction and construction work including preparation of the mix design(s), quality control, quality assurance, surface preparation; installation of wick drains, weep holes and pipes; supply and installation of the shotcrete; preconstruction and construction testing, curing; the supply, transportation and testing of shotcrete panels; any core extraction of the in-place shotcrete, and any other testing to determine curing or protection requirements.

209.02 Quality Management

209.02.01 General - The Contractor shall provide on-site quality control and independent quality assurance for shotcrete operations.

209.02.02 Qualifications - The Contractor's QC representative shall have designed shotcrete for 3 projects over the last 5 years. The Contractor's QC representative conducting onsite shotcrete tests shall be an ACI or CSA certified concrete technician.

209.02.03 Specific Requirements - For wet and dry process shotcrete, quality control shall include, but not be limited to the following:

- Observation of surface preparation before shotcrete application.
- Observation and testing during shotcrete application.
- Observation of curing operations.

209.02.04 Other Requirements - The Contractor's QC representative shall be present at other times as are appropriate to assist the Contractor in the preparation, assessment and adjustment of the various procedures to be used for shotcrete operations.

209.02.05 Field Report - The Contractor' QC representative shall prepare a field report for each site visit, signed by the QC representative. The report shall include details of the progress of shotcrete operations, any recommendations made to the Contractor and any problems encountered by the Contractor.

209.02.05 Quality Assurance - The Contractor shall perform QA in accordance with the Contract, to assess the quality of the product, process and the effectiveness of the QC.

209.03 Submittals

209.03.01 The Contractor' Quality Management Plan shall include the following documentation;:

a) Qualifications and Experience of the Contractor - Provide a statement of the qualifications, experience and work function of all personnel assigned to shotcrete duties. A statement of previous work experience on similar projects shall also be provided. This statement shall include the project name, location, type of shotcrete, volume of shotcrete, year constructed and the owner contact name. The company, the nozzle operator and the predampener/pot operator shall each have a minimum of 5 years demonstrated experience on shotcrete work.

b) Qualifications and Experience of the Contractor's QC representative - The qualifications shall include the name of the consulting firm, and the name and qualifications of the Consultant's on-site representative who will be providing the quality control. The following information shall be included in the qualifications submitted:

- Project name, location and experience.
- Name and phone number of the owner contact who can verify the experience of the Contractor's Consultant's site representative.

c) Proposed Laboratory Testing Agency - Provide a statement of the testing agency's experience in performing laboratory tests on shotcrete. The agency shall be an independent, qualified laboratory with a proven record of performing tests on shotcrete on at least 5 previous projects. The testing agency shall be certified to CSA A283.

d) Certification of the Concrete Technician - Provide evidence of ACI or CSA Certification for the concrete technician who will be providing on site shotcrete tests.

209.03.02 The Contractor shall submit for review the following documentation at least 10 days prior to commencing shotcrete operations:

a) Materials Records - Provide records showing source and proof of conformance to project specifications of the following materials:

- Portland Cement (provide mill certificates)
- Silica Fume (provide mill certificates)
- Mix Water
- Aggregates
- Admixtures (provide supplier data sheets)
- Fibre Reinforcement (if specified), include load/deflection curves to ASTM C 1018 for the proposed mix design
- Mesh Reinforcement (if specified, provide mill certificates)

b) Previous Performance Data - Provide previous performance data for the proposed shotcrete mix design, if available.

c) Proposed Shotcrete Mix Design (Proportions) - Provide documentation for the proposed shotcrete mix design. Include such back-up data as requested by the Ministry Representative. The proposed mix design submittal shall include:

- Mix design number.
- Batch quantities in kg/m³ based on aggregates in an SSD (Saturated Surface Dry) moisture state for wet process shotcrete and ready mix supplied dry process shotcrete. Batch quantities for dry bagged supply shall be based on mass of aggregates in a dry state.
- A total cementitious content of not less than 20% by mass of the total ingredients.
- Aggregate source, bulk density, absorption, combined gradation data and a grain size distribution graph scaled between 0.075 mm and 75 mm showing the percent passing curve for the combined aggregate.

d) Proposed Shotcrete Imprint - Provide a drawing of the proposed shotcrete imprint including content, imprinting method and materials that will be used to produce the imprint.

209.03.03 The Contractor shall submit for information the following documentation within 1 day following each day of shotcrete application:

a) Field Report - Provide original field reports signed by the Contractor's QC representative.

b) Daily Records - Provide daily shotcrete records with the following information (certified by the Contractor's QC representative in the case of wet process shotcrete):

- Contractor Name
- Nozzleman Name
- Helper Name
- Contractor's QC Representatives Name (for wet process shotcrete)
- Date
- Weather
- Contractor's start and end time
- Shotcrete equipment type
- Temperature
- Type of shotcrete (wet or dry process)
- When is slump measured, before or after admixture and fibres

Submit the following information for each batch of shotcrete:

- Batch number
- Location of shotcrete
- Mix design number
- Batch time at plant (wet process)
- Batch arrival time on site
- Truck departure time (wet process)
- Placement start time
- Placement end time
- Quantity used
- Water/cementitious materials ratio (wet process)
- As-shot air content (wet process)
- Slump (wet process) - indicate when measurement is made (e.g.: after air entrainment is added, before superplasticizer is added and before fibre reinforcement is added)
- Percent rebound (visual estimate)
- Average thickness of shotcrete
- Test panel number
- Shotcrete delivery tickets from batch plant (wet process)
- Comments on the quality of the shotcrete and any problems that arise
- Proposed revised mix design

209.03.04 The Contractor shall submit the laboratory test results within 2 working days after testing. These test results shall compare the shotcrete performance with the requirements given in Table 209-A. Records shall include all reporting requirements outlined for construction testing. Records shall also include an original summary report of all test results, showing the sample number, sample origin, panel number, panel origin, date the panel was sprayed, panel shipping date, date panel received at the laboratory, photos of prepared test samples, test date, sample age at testing, test results for each sample, average test results of the three samples for each test at a given age, and

acceptability of the results.

209.04 Definitions - The following definitions refer to words and terms used in this specification. For definitions not covered in this document, refer to ACI 506R and ACI 506.2.

209.04.01 Blow pipe - Air / water jet operated by nozzle operator's helper during shotcrete placement to assist in keeping rebound and overspray out of the work.

209.04.02 Dry Process Shotcrete - A process where most of the water added to the shotcrete mixture is added at the nozzle.

209.04.03 Overspray - Shotcrete material deposited away from the intended receiving surface.

209.04.04 Pot (Sometimes called Gun) - Machine used to meter the shotcrete into the hose.

209.04.05 Predampener - Device used to mix a portion of the mixing water with the shotcrete. Can also be used to mix accelerator with shotcrete.

209.04.06 Saturated Surface Dry - The state of a material in which it will neither add nor subtract moisture from other material placed in contact with it.

209.04.07 Slake - A process of deterioration of freshly exposed rock due to exposure to the atmosphere and/or water.

209.04.08 Sloughing (also called sagging) - Subsidence of shotcrete, due generally to excessive water in the mix or placing too great a thickness of shotcrete in a single pass.

209.04.09 Weep Pipes - Plastic pipes inserted in water bearing fissures to relieve water pressure from behind the shotcrete.

209.04.10 Wet Process Shotcrete - Shotcrete in which all of the ingredients are mixed before introduction into the delivery hose. Compressed air is introduced to the material flow at the nozzle. If an accelerator is used, it is normally added at the nozzle.

MATERIALS

209.11 Materials

209.11.01 Cement - Cement shall be Type 10 or 30.

209.11.02 Silica Fume - Silica fume shall meet the requirements of CAN/CSA-A23.5-M, Supplementary Cementing Materials, Type U Silica Fume, with the following additional requirements:

- Minimum $Si0_2$ content - 90 percent by mass
- Maximum carbon content: 5 percent by mass

209.11.03 Pozzolan - Pozzolan shall be Type F flyash.

209.11.04 Water - Surface or ground water may be used for cleaning the slope and curing provided it is free of deleterious substances. Verification that the water intended for use is acceptable shall be made by the Designer and Quality Manager. **209.11.05 Aggregates -** Aggregate shall be normal density. The combined aggregate gradation limits shall meet the limits shown in Table 209-B.

TABLE 209-A SHOTCRETE PERFORMANCE REQUIREMENTS

TEST DESCRIPTION	TEST METHOD	AGE (DAYS)	SPECIFIED REQUIREMENT
*Maximum Water / Cementitious Materials Ratio		-	0.45
*Air Content - As Shot, %	CAN/CSA A23.2-4C	-	4 ± 1½
*Slump at discharge into shotcrete pump, mm	CAN/CSA A23.2-5C	-	80 ± 30
Minimum Compressive Strength, MPa	CAN/CSA A23.2-14C	7 28	30 40
Minimum Flexural Strength, MPa	ASTM C 1018	7	4
Minimum Toughness Performance Level (TPL)	ASTM C 1018 and this specification	7	III
Maximum Boiled Absorption, % Maximum Volume of Permeable Voids, %	ASTM C 642	7 7	8 17

Notes: * Properties apply to wet process shotcrete only.

209.11.06 Admixtures

a) General - Chemical admixtures shall include air-entraining, water reducing and set retarding admixtures where appropriate.

b) Superplasticizers - Superplasticizers shall meet the requirements of CAN3-A266.6-M.

c) Accelerating Admixtures in General - Accelerating admixtures shall only be used with the written consent of the Designer

d) Accelerating Admixture Requirements - Accelerating admixtures:

i) shall be non-chloride based accelerating admixtures meeting the requirements of CAN/CSA A23.1M Clause 15.1.7.1;

ii) shall be compatible with the cement being used when tested in accordance with ASTM C 1117;

iii) shall have a demonstrated ability, when used at the required proportions, to produce accelerated shotcrete meeting the performance criteria specified in Table 209-A;

iv) for wet process shotcrete shall be liquid, dispensed at the nozzle in closely controlled quantities;

v) for dry process shotcrete shall be liquid, dispensed at the nozzle in closely controlled quantities; or be powdered and either:

- integrally mixed in the case of dry bagged material or;
- mechanically proportioned through the predampener in the case of ready mix supply; metering of accelerators directly into the pot will not be permitted.

TABLE 209-B COMBINED GRADATION LIMITS FOR SHOTCRETE AGGREGATE

U.S. STD SQUARE MESH	METRIC SIEVE SIZE	TOTAL PASSING EACH SIEVE SIZE, % BY MASS
3/4	20 mm	
1/2	14 mm	100
3/8	10 mm	90-100
#4	5 mm	70-85
#8	2.5 mm	50-70
#16	1.25 mm	35-55
#30	0.630 mm	20-35
#50	0.315 mm	8-20
#100	0.160 mm	2-10

209.11.07 Fibre Reinforcement - Unless otherwise specified, shotcrete reinforcement shall consist of steel or polyolefin fibres. The fibres shall be capable of meeting the performance criteria specified in Table 209-A and the following requirements:

a) Steel Fibres - These fibres shall meet the requirements of ASTM C 1116 Type I.

b) Polyolefin Fibres - These fibres shall only be used in wet process shotcrete and shall meet the requirements of ASTM C 1116 Type III.

209.11.08 Mesh Reinforcement - Where specified instead of or in addition to fibre reinforcement, shotcrete shall be reinforced with mesh anchored to the surface before shotcrete application. For fibre reinforced shotcrete, additional reinforcement may be required by placing mesh panels up to 0.60 m by 0.60 m without anchors between successive shotcrete lifts where final shotcrete thicknesses are expected to exceed 300 mm in local areas.

a) Mesh Materials - Mesh shall consist of welded wire fabric 102 mm by 102 mm opening (4 by 4 inch) by MW13.3 / MW13.3 (8 gauge) wire meeting CSA G30.5 and hot dip galvanized meeting ASTM A 123.

b) Mesh Anchor Materials - Anchors shall consist of ASTM A 615 Grade 400 steel, minimum 19 mm nominal diameter. Anchorage may be resin or grout. Grout anchorage may be Ocean "Microsil Anchor Grout" or Target "Unsanded Silica Fume Grout". Resin anchorage may be Celtite Lokset or Ground Control cartridges. Face plates shall be 150 mm by 150 mm by 6 mm. Face plates, nuts and washers shall be capable of developing full anchor pullout capacity.

c) Mesh Coverage and Overlap - Mesh shall be installed over the entire area to be shotcreted, unless otherwise specified. Mesh joints shall be lapped 1.5 mesh openings in each direction (horizontally and vertically) and tied at minimum 100 mm intervals with 16 gauge tie wire to achieve full contact between mesh layers.

d) Mesh Anchor Spacing - Mesh shall be anchored on minimum 1.2 m centres in each direction or closer where required to form the mesh to within 50 mm of the slope surface. The mesh shall be in full contact with the surface where practicable. The mesh may also be anchored to any previously installed rock bolts provided the spacing is consistent with the spacing required for the above anchors.

e) Anchor Installation - Anchors shall be

installed as recommended by the anchor manufacturer. The minimum anchor embedment length shall be 1.0 m. Resin or grout shall be applied to full anchor embedment.

f) Mesh Anchor Testing - A total of 5% of the anchors shall be load tested to 20 kN. This load shall be held for 10 minutes. A load loss of greater than 10% of the applied load shall be indicative of anchorage failure. Any failed anchors shall be replaced.

209.11.09 Anchor Reinforcement - Where specified, additional anchor reinforcement shall be installed. Anchor materials, installation and testing shall meet the requirements for mesh anchors unless otherwise specified. Face plates for these anchors shall be "Spider Plates". These anchors and plates shall be fully embedded in the shotcrete.

209.11.10 Materials Handling and Storage

a) General - All materials shall be stored and handled in a manner that will prevent damage, deterioration or contamination.

b) Hot Weather Storage - During hot weather prebagged material shall be stored in a manner that will allow maximum cooling of the material.

c) Cold Weather Storage - During cold weather prebagged material shall be stored in a manner that will provide maximum heat retention of the material.

209.12 Performance Requirements

209.12.01 General - Shotcrete shall meet the performance requirements in Table 209-A.

209.12.02 Air Content - The air content shall be determined on wet process shotcrete sprayed into a CAN/CSA-A23.2-4C air pressure meter base or freshly applied material removed from in-place and consolidated into the air meter base by rodding.

CONSTRUCTION

209.31 Batching, Mixing and Conveyance

209.31.01 General - Shotcrete shall be batched, mixed and supplied using (one of) the following system(s):

a) dry bagged pre mix supply to ASTM C 928 or:

b) ready mix supply

209.31.02 Mixing - Shotcrete supplied shall be uniformly mixed with no evidence of segregation or improper mixing.

209.31.03 Batching and Conveyance - Ready mix supplied shotcrete shall be batched and conveyed at a rate that allows a consistent application of shotcrete. Irregularities of conveyance that result in delays or a surplus of aged material on site shall be avoided. Shotcrete that has stiffened excessively or is more than 90 minutes old from the time of batching will not be accepted.

209.32 Preconstruction Testing

209.32.01 General - The Contractor shall perform preconstruction shotcrete tests (prior to full-scale shotcrete operations), at a shotcrete test site , to demonstrate the competence of the crew, the adequacy of the equipment and the adequacy of the shotcrete mix design. The requirement for preconstruction testing may be waived by the Ministry Representative if sufficient documentation of previous satisfactory performance on similar work is provided.

209.32.02 Previous Performance - Documentation for previous satisfactory performance shall meet the following requirements:

a) Company - The company shall have applied more than 100 m³ of shotcrete on a minimum of 3 projects in the last 5 years.

b) Nozzle Operator - The nozzle operator shall have applied more than 100 m³ of shotcrete on a minimum of 3 projects in the last 5 years.

c) Predampener/Pot Operator - The predampener/pot operator shall have mixed more than 100 m³ of shotcrete on a minimum of 3 projects in the last 5 years.

d) Contacts - The contact name and number for each project above shall be supplied upon request.

209.32.03 Application - Shotcrete applied at the shotcrete test site shall be applied in accordance with the general shotcrete provisions in these specifications.

209.32.04 Test Panel - One test panel shall be gunned and tested for the preconstruction shotcrete testing as described for construction testing.

209.32.05 Test Results - The Contractor shall not apply shotcrete outside of the shotcrete test site area until the preconstruction shotcrete test results have demonstrated that the shotcrete system employed is satisfactory and the test site information has been

supplied to the Ministry Representative and the test site information has been signed off by the Designer.

209.32.06 Revised Methods - If the results of the shotcrete tests are unsatisfactory, the Contractor shall adopt such revised methods as are necessary to achieve the required results.

209.32.07 Additional Tests - The Contractor shall perform, at any time during the progress of the work, additional shotcrete tests at designated test sites if the shotcrete applied in the areas specified in this Contract is unsatisfactory in the opinion of the Ministry Representative.

209.33 Weather Conditions

209.33.01 Heavy Rain or High Wind - During periods of heavy rain or high wind, the Contractor shall ensure that the quality of the finished shotcrete product is not compromised. Shotcrete shall not be applied during such weather unless protective measures can be practicably taken.

209.33.02 Hot Weather - During periods of hot weather, when ambient temperatures are expected to rise over 30°C, the following requirements shall be met:

a) Curing - The Contractor shall implement curing immediately after the shotcrete has been placed.

b) Temperature at Placement - The Contractor shall take any steps necessary to ensure that the temperature of the shotcrete supplied to the site is less than 25°C at the time of placement.

209.33.03 Cold Weather - During periods of cold weather, where the minimum ambient air temperature falls below 5°C, the following requirements shall be met:

a) Temperature at Placement - The Contractor shall take whatever steps are necessary to ensure that the in-place shotcrete temperature is between 10 and 25°C at the time of placement.

b) Initial Curing Temperature - The Contractor shall ensure that, during the initial curing period (4 days), the in-place shotcrete temperature will not fall below 5°C. Shotcrete shall not be placed unless measures can be practicably taken to ensure these requirements are met.

c) Temperature at the Pot - The Contractor shall not allow the temperature of the dampened dry process shotcrete at the pot to exceed 30°C.

d) Ice, Snow or Frozen Surfaces - Shotcrete shall not be applied to frozen surfaces, or surfaces covered with frost, ice or snow.

209.34 Surface Preparation - Surfaces to which shotcrete is applied shall be cleaned of loose rock, dirt and debris which if left in place could impair the bond of shotcrete to rock, concrete or existing shotcrete. Water or air and water jets shall be used to clean rock surfaces. Compressed air only shall be used to clean rock surfaces that have a tendency to slake from contact with water.

209.35 Drainage - Unless otherwise controlled, water seepage from slopes shall be controlled by installing weep holes, weep pipes, and wick drains. Surface runoff shall be controlled using suitable water diversion systems.

209.35.01 Weep Holes - Weep holes (and weep pipes as required) shall be installed before shotcrete is applied. These weep holes shall be on 1 to 1.5 m centres in the areas to be shotcreted in joints, fractures and faults, or at a closer spacing where required by the Designer. Holes and pipes shall be inclined to permit free drainage.

209.35.02 Weep Pipes - To prevent blockage of weep holes due to entry of shotcrete or rebound materials, weep pipes shall be secured in weep holes and temporarily plugged at the outer end of the pipe. The plugs shall be removed when shotcreting is completed. Weep pipes shall consist of 25 mm minimum outside diameter Schedule 40 plastic pipe.

209.35.03 Active Seepage Areas - Where active seepage occurs, weep holes shall be drilled approximately 150 mm or more to intercept fractures or openings wherever practicable and weep pipes shall be installed prior to the application of shotcrete. Where drilling is not practicable, the weep pipes shall be secured in the fractures by other appropriate means.

209.35.04 No Active Seepage Areas - Where no active seepage occurs, weep holes shall be installed in joints, fractures and faults by one of the following methods:

a) by securing the weep pipes by appropriate means before shotcreting;

b) by marking locations using sticks or other appropriate means before or during shotcreting and drilling at the marked locations after shotcreting. Weep holes shall be 25 mm diameter and shall be drilled to intersect the joints, fractures and faults under the shotcrete at the marked locations. Weep pipes are not required for such holes.

209.35.05 Weep Pipe Removal or Cutting - Weep pipes shall be removed or cut level with the shotcrete surface after the shotcrete has reached initial set, within one day after application.

209.35.06 Additional Weep Holes - Additional weep holes shall be drilled after shotcreting where there are visible signs of water pressure, such as seepage and damp spots in the hardened shotcrete.

209.35.07 Wick Drains - In areas of active seepage, wick drains shall be installed before shotcreting in addition to weep holes if requested by the Designer. Wick drains shall consist of a flexible polypropylene core drain wrapped with a proven durable geotextile filter fabric. Wick drains shall be 3 mm by 100 mm size (variable length), with #120 sieve filter, 0.17 mm/s filter permeability, and 0.8 kN core grab tensile strength. They shall be able to withstand all handling, abrasion and distortion that occur during installation. Wick drains shall be secured to the surface on maximum 0.5 m centres with pneumatically driven nails, or authorized substitute fasteners. Fastener spacing shall be reduced where required to ensure wick drains are in full contact with the surface. Wick drains shall be installed and covered with shotcrete in a manner that allows gravity drainage to the toe of the slope.

209.35.08 Surface Runoff - Where the bond of shotcrete to rock may be impaired by surface water runoff, this water shall be controlled and directed away from the area to be shotcreted.

209.36 Existing Rock Bolts - Within or near areas designated for shotcrete where rock bolts were installed (not for the sole purpose of securing mesh) prior to shotcrete application, the rock bolts shall be protected or extended through the shotcrete as follows:

209.36.01 Extension of Rock Bolts - Where rock is fractured within 0.5 m of the existing rock bolts, the face plates and nuts shall be removed before shotcrete is applied. The bar shall be extended, where necessary, using a short piece of bar and a coupler. The shotcrete shall be applied up to and against the existing bar. The face plates and nuts shall then be reinstalled over the shotcrete after the shotcrete has cured. Each face plate shall be replaced to the same bar from which it was removed. The exposed bar length shall be 100 mm following reinstallation of the face plate. The bar shall be nominally tensioned (using a wrench to the nut).

209.36.02 Protection of Rock Bolts - All other rock bolts shall be protected from overspray. Where overspray was not anticipated, the affected rock bolts shall be cleaned off by appropriate means.

209.37 Application

209.37.01 General - The Contractor shall ensure that the shotcrete operations are conducted in a satisfactory manner in accordance with these specifications.

209.37.02 Wetted Surface - Not more than one hour prior to application of shotcrete, all surfaces (except slaking ground) to be shotcreted shall be sprayed with water. Wetted surfaces shall be allowed to dry back to a saturated-surface-dry condition prior to application of shotcrete. If necessary, a blow-pipe shall be used to facilitate removal of surface water. Compressed air used in the blow pipe shall be oil-free.

209.37.03 Shotcrete Layers - Wherever possible, shotcrete shall be applied to the full thickness in a single layer, provided it does not slough or become delaminated. If multiple lifts are required, the previous shotcrete lifts shall be scraped or broomed prior to the time of set to remove any loose material, rebound, overspray, laitance or any other material that may impair bond of subsequent layers. Previous lifts shall be prevented from drying out by moist curing. Construction joints shall be to a 45° edge.

209.37.04 Rebound and Overspray - A blow pipe shall be used during the course of the work to remove accumulations of rebound and overspray from areas to be shotcreted. Rebound and overspray shall not be incorporated in the completed work.

209.37.05 Thickness and Area Covered - In general, an average thickness of 100 mm of shotcrete shall be applied over fractures, joints and faults in the areas designated for shotcrete, unless the Designer requires a greater thickness to ensure performance.. In general, shotcrete shall cover laterally to a maximum of 0.50 m beyond fractures, joints and faults over solid rock or concrete areas.

209.37.06 Access - Suitable devices shall be employed to allow access to the work for shotcreting. Access shall be such that fresh shotcrete is not damaged by hoses, equipment or personnel.

209.37.07 Surface Finish - The shotcrete surface shall be left in a natural gun state unless otherwise specified.

209.37.08 Embedments - A cover of 50 mm of shotcrete shall be applied over any reinforcing steel or other embedments. If greater total thicknesses of shotcrete are applied to encapsulate the embedments this work shall be performed at no cost to the Ministry.

209.37.09 Imprint - The plastic shotcrete surface shall be neatly imprinted with capitalized lettering 75 mm high with the following information:

- MoT
- (the shotcrete completion date, in the format MMM, YYYY)
- (the name of the Contractor)

The imprint shall be located in a small area in the bottom right corner of each continuous shotcrete area. Where practicable, the information shall be readable unaided from highway level following shotcrete curing. Imprints produced by scraping letters into the plastic shotcrete will not be acceptable.

209.38 Curing and Protection

209.38.01 Curing Compounds - Membrane curing compounds shall not be used

209.38.02 Moist Curing - The surface of the shotcrete shall be maintained in a moist condition for a minimum period of 4 days following shotcrete application or until an in-place compressive strength of 20 MPa has been reached. Wet burlap and polyethylene sheet or water sprays shall be used to provide moist curing.

209.38.03 Protection from Freezing - The shotcrete shall be protected from freezing for a minimum period of 7 days after placing or until an in-place compressive strength of 20 MPa has been reached, whichever occurs first.

209.38.04 Sampling and Testing - If compressive strength samples are used to determine curing or protection requirements, the cost for obtaining the samples, shipping the samples and performing the tests shall be the responsibility of the Contractor. Extraction of cores shall be in accordance with CAN/CSA A23.2-14C. Compressive strength testing shall be performed according to requirements for construction testing.

209.39 Construction Testing

209.39.01 On-Site Tests - For wet process shotcrete, the following on-site tests shall be conducted for each truck:

- As-placed air content.
- Slump at discharge into pump.

209.39.02 Thickness Monitoring - The Contractor shall monitor the thickness of shotcrete applied using depth gauges or probes. Thickness shall be measured on 1.0 m centres or closer as required to provide thickness control.

209.39.03 Test Panels in General - The Contractor shall produce test panels in accordance with the requirements of ASTM C 1140, but panels shall have minimum base dimensions of 500 mm x 500 mm and a minimum depth of 125 mm. The form base shall be made from 19 mm thick sealed plywood, and side pieces shall be made from two by six timbers. The test panel molds shall have 45° sloped sides to permit escape of rebound.

209.39.04 Test Panel Frequency - One construction test panel shall be cast at the start of each day of shotcrete placement and another after each 50 m^3 of shotcrete being placed that day,.

209.39.05 Test Panel Gunning and Marking - Test panels shall be gunned using the same nozzle and length of hose as the production shotcrete. The test panels shall be gunned at the location of the in-situ shotcrete. Construction test panels shall contain no reinforcement or embedments (other than fibre reinforcement). For each test panel the Contractor shall provide a record of the test panel number, date cast, date shipped and date received by the laboratory.

209.39.06 Test Panel Curing and Transportation - Test panels shall be field cured, in the forms, in the same manner as the specified shotcrete work, for a minimum period of 48 hours, prior to transport to the test laboratory. Panels shall be covered with wet burlap and plastic sheet to prevent drying. In cold weather, panels shall be protected from temperatures below 5°C by covering with thermal insulation blankets or by placing panels in a suitably heated enclosure. Test panels shall be transported in their forms and kept moist, with care being taken not to crack or damage the shotcrete.

209.39.07 Test Panels in Laboratory - The test panels shall be placed in a moist room in the laboratory, maintained at a temperature of 23 ± 2°C and relative humidity of 98 ± 2%. After age 3 days the test panels shall be removed from the forms and returned to the moist room until the time of testing.

209.39.08 Test Specimen Extraction and Storing - At specified test ages, test specimens shall be extracted from the panels using either diamond sawing or coring. Test specimens shall be maintained in a moist state in the moist room in the laboratory or in a lime-saturated water bath maintained at 23 ± 2°C until the time of testing.

209.39.09 Compressive Strength Test Specimens - Compressive strength test specimens shall preferably be 75 mm diameter cores. Length/diameter ratios shall not exceed 2:1 and shall not be less than 1:1.

209.39.10 Compressive Strength Testing - For each panel, three specimens shall be tested for compressive strength at each specified age. Compressive strength tests shall be conducted in accordance with CSA/CAN-

A23.2-14C. Measured compressive strengths shall be corrected to equivalent 2:1 length:diameter cores, using the core correction factors given in CSA/CAN-A23.2-14C (Table 1). Reporting of results shall be according to Section 5 (Reporting) in CSA/CAN-A23.2-14C.

209.39.11 Flexural Strength and Toughness Test Specimens - Flexural strength test specimens shall have dimensions of 100 mm x 100 mm x 350 mm. Reporting of results shall be according to "11. Report" in ASTM C 1018 and shall include all load - deflection graphs.

209.39.12 Flexural Strength and Toughness Testing - For each panel, three specimens shall be tested for flexural strength and toughness at the specified age. Flexural strength and toughness tests shall be conducted in accordance with ASTM C 1018 with the following additional requirements:

a) Measuring System - A "Japanese Yoke" or other measuring system that precludes any extraneous deflection measurements shall be used. An automatic data acquisition system or equivalent shall also be used during the tests.

b) Accuracy of Measurement - The accuracy of the first crack deflection shall be verified by comparing the measured value with the theoretically calculated value by using the formula:

$$\delta = \frac{23\,PL^3}{1296\,EI} \times \left[1 + \frac{216\,H^2(1+\mu)}{115\,L^2}\right]$$

Where:

δ = mid-span deflection in mm
P = load at first crack in kN
L = load span in mm
E = modulus of elasticity (for convenience usually based on an estimated compression modulus of elasticity) in GPa
I = moment of inertia in mm^4
H = height of specimen in mm
μ = Poisson's Ratio (for convenience usually based on an assumed compression Poisson's Ratio of 0.2)

For a 100x100x350 mm beam tested on a 300 mm span the first crack deflection for typical fibre reinforced shotcrete will have first crack deflection in the range of about 0.03 to 0.05 mm. Extraneous deflections have been eliminated if first crack deflections are within this range.

c) Recording of Deflections - An LVDT connected to an x-y plotter or digital recording shall be used to capture load vs. deflection response. Dial gauges shall not be used for recording deflections. Bending Load shall be plotted against Net Midspan Deflection.

d) Testing Machine - A deflection-control testing machine shall be used, but shall not be of the closed-loop or soft testing type.

e) Total Deflection - Deflection measurements shall continue until a total deflection of 2 mm is reached.

209.39.13 Toughness Performance Level Data Analysis - Toughness Performance Level of flexural strength specimens shall be obtained by the following analysis:

a) Template - A template shall be created using the percent of design flexural strength shown in Table 209-C, but expressed in load equivalent units. The following formula shall be used:

$$P_d = \frac{\sigma \times B \times H^2}{L} \times \frac{1}{1000}$$

to determine load equivalent units, where:

P_d = design flexural load in kN
σ = design flexural strength in MPa
B = beam width in mm
H = beam depth in mm
L = load span in mm

TABLE 209-C TOUGHNESS PERFORMANCE LEVEL

TOUGHNESS PERFORMANCE LEVEL	RESIDUAL FLEXURAL STRENGTH BY % OF DESIGN FLEXURAL STRENGTH	
	1/600 Span (0.5 mm)	1/150 Span (2.0 mm)
III	50%	30%

b) Master Template - A master template shall be produced to the same scale as the load vs. deflection data recorded from the toughness test. The master template shall be superimposed over the actual load deflection curves and compared with the recorded data.

c) Acceptability Criterion - To conform to a given Toughness Performance Level the recorded data shall meet the specified flexural strength and fall above the line representing the given

Toughness Performance Level at both the 0.5 mm and 2 mm net midspan deflection.

209.39.14 Boiled Absorption and Permeable Voids Test Specimens - Specimens for boiled absorption and permeable voids testing shall be 75 mm cubes cut from the broken ends of flexural test prisms, or extracted 75 mm diameter cores at least 100 mm long.

209.39.15 Boiled Absorption and Permeable Voids Testing - For each panel, three specimens shall be tested for boiled absorption and permeable voids at the specified age. Boiled absorption and permeable voids tests shall be conducted in accordance with ASTM C 642. The report of results shall include all information recorded for 6.1 Calculation in ASTM C 642.

209.39.16 Laboratory Test Results - For each type of test, test results shall be considered acceptable if the average of three (3) specimens tested at a given age meets or exceeds the criteria specified in Table 209-A. The test results for an individual specimen shall not be less than 80% of the criteria specified for compressive strength, and not more than 120% of the criteria specified for boiled absorption and permeable voids.

209.40 Shotcrete Deficiencies and Repair

209.40.01 General - The Contractor shall, whenever possible, correct deficiencies while the shotcrete is still plastic. For any proposed changes to the initial mix design, a solid volume calculation shall be performed and submitted upon request. The calculation shall include the mass, density and volume of each constituent material, the total yield, and a comparison between measured (ASTM C 138) and the calculated plastic density.

209.40.02 Deficiency Types - Shotcrete deficiencies other than failure to meet the performance requirements include, but are not limited to, the following:

a) Improper consolidation of the shotcrete because of incorrect spraying angles or improper distance of the nozzle from the receiving surface.

b) Improper control and removal of overspray and rebound material.

c) Excessive shotcrete or fibre rebound.

d) Any evidence of excessive plastic or drying shrinkage cracking.

e) Segregation, incorporation of sand lenses, excessive voids, tears, sags, sloughs or delaminations in the work.

f) Inadequate thickness.

209.40.03 Shotcrete Repair - Shotcrete with deficiencies as defined above shall be removed and replaced to these specifications.

209.41 Shotcrete Evaluation – The Designer and Quality Manager will evaluate and report on the quality of the in place shotcrete. The Ministry Representative will audit in-place shotcrete.

209.41.01 Core Extraction - Cores shall be extracted from the in-place shotcrete in accordance with CAN/CSA A23.2-14C, at locations detailed by the Quality Manager, and tested for compliance to the project specifications. Six (6) cores shall be extracted from every 200 m² of shotcreted surface.

Up to an additional 2 cores per 200 m^2 , from locations chosen by the Ministry Representative, shall be extracted by the Contractor for the Ministry's quality audit program.

209.41.02 Non-Destructive Testing - The shotcrete will be checked for delaminations, voids or other deficiencies using appropriate non-destructive testing procedures.

DAILY SHOTCRETE RECORD

File #: _____

PROJECT NAME_____

PROJECT NO._____

(TO BE SUBMITTED WITHIN 1 DAY AFTER EACH DAY'S OPERATION)

Contractor Name _____ Date _____
Nozzleman Name _____ Weather _____
Helper Name _____ Temperature_____°C
Contractor's Start _____End _____Time
Contractor's QC Representative _____ Shotcrete Equipment Type_____

Shotcrete Type: (circle) WET DRY Process

For wet process, when is slump measured (circle) BEFORE AFTER Air entrainment added
BEFORE AFTER Superplasticizer added
BEFORE AFTER Fibre Reinforcement added

On-Site Details and Tests for Shotcrete:

Batch #	Location		Mix Design #	Batch Times:			Spray Start Time	Spray End Time	Quantity Used m^3	Water/ Cement Ratio	Sample (As-Shot) Air Content	Slump (wet mix) (mm)	Re-bound est. (%)	Average Thickness of In-Place Shotcrete (mm)	Test Panel No.
	Zone	Area		From Plant	On Site From	To									

Comments: _____
_____
_____

Certified Correct:_____ _____
(Contractor's QC Representative) (Title)

(Use Additional Sheets if Necessary)

SECTION 211

PORTLAND CEMENT CONCRETE

Note: For any SS Drawing (SP XXX - XX) referenced in this Section refer to the like-numbered Drawings in the "Standard Specifications For Highway Construction" 2006 edition.

211.01 General. This Section describes the requirements for Portland Cement Concrete. Requirements include quality control and quality assurance, constituent materials, mix design proportioning, sampling and testing of the materials and concrete, batching, mixing, transporting, placement, finishing and curing. The Contractor shall supply all necessary materials.

TABLE 211-A APPLICABLE STANDARDS

CAN/CSA -A23.1	Concrete materials and methods of concrete construction
CAN/CSA -A23.2	Methods of test and standard practices for concrete
CAN/CSA -A3000	Cementitious Materials Compendium
CAN/CSA -S269.3	Concrete Formwork
CAN/CSA -A283	Qualification Code for Concrete Testing Laboratories

211.02 Applicable Standards. The standards listed in Table 211-A shall apply unless specified otherwise herein . All referenced standards shall be to the current editions at time of Award.

211.03 Concrete Mix Designs and Submission Requirements.

211.03.01 General. The Contractor shall be responsible for quality control and quality assurance of all components of the concrete operation, including but not limited to, aggregate and component quality, proportioning, test batching, batching, mixing, transporting, placing, consolidating, finishing, curing, and all necessary quality control, quality assurance and verification testing of the components and the fresh and hardened concrete.

211.03.02 Proportioning. The Designer shall be responsible for proportioning and designing all concrete in full compliance with the stated mix specifications. All mix designs shall use the "Absolute-Volume Method" for mix proportioning.

211.03.03 Alkali-Aggregate Reactivity (AAR) Preventative Requirements. Contrary to CSA A23.2 27A, the total alkali content (Na_2O equivalent) contributed by the Portland cement to the concrete mix shall not exceed 2.0 kg/m^3. The Ministry Representative reserves the right to alter this limit or to redefine the concrete mix design parameters based on the review of the mix design submittal.

211.03.04 Mix Design Submittals. The Designer shall

submit a report outlining the proposed mix design for each classification of concrete to the Ministry Representative for review at least 2 weeks in advance of when concrete production is scheduled to commence. Review of the mix design by the Ministry does not constitute acceptance of the concrete. Acceptance of the concrete will be based on the test results and the performance and quality of the concrete and concrete components placed on the project. No concrete shall be placed prior to receiving Ministry review comments of the mix design. Each mix design submittal shall include all of the following:

Project Specifics

- Name and location of the proposed supplier.
- Distance and expected travel time from batch plant location to Site.
- Expected method of batching, transporting and placing concrete.
- Specified mix parameter requirements.
- Name and contact information of certified quality control testing laboratory and certified testing personnel. Certification shall be in accordance with the requirements of CSA A23.1.
- Name and contact information of certified quality assurance testing laboratory and certified testing personnel. Certification shall be in accordance with the requirements of CSA A23.1.

Materials

All materials shall comply with the requirements as listed in DBSS 211.04 .

- Cementitious Materials – Types and source of each material including mill test reports and manufacturer's certificates of compliance.
- Aggregates – Type and source of all individual aggregate products including individual gradations and all other aggregate quality testing as described in Table 211-D for each product and each aggregate source.
- Admixtures – Type and source of all admixtures,

including expected dosage rates, point of addition to the mix, compatibility documentation, and individual technical data sheets.

- Water – The source of mixing water. (depending on the source, the Ministry Representative may request additional test documentation).
- Miscellaneous – Documentation for all other materials proposed for the mix, showing conformance with applicable Ministry requirements and manufacturer/industry guidelines and standards.

Proportioning

- The concrete proportions expressed in terms of quantity of each component.
- Fresh and hardened concrete properties.
- The mass of materials in kilograms (kg) and the absolute volume in cubic metre (m^3) contributed by each material in the mixture.
- The quantity of each individual cementitious material added in a powder state shall be expressed as kilograms per cubic metre (kg/m^3).
- Pozzolans that are added in slurry shall have their respective solid and water contents expressed as kilograms per cubic metre (kg/m^3).
- The quantity of each separately batched size of coarse aggregate and fine aggregate shall be expressed as kilograms per cubic metre in a saturated-surface-dry (SSD) condition.
- The quantity of water shall be the maximum amount allowed based on the maximum specified water-to-cementitious material ratio (W/C_m), corrected to account for any additional water that may be contributed by other materials used in the proposed mix.
- The quantity of water shall be expressed as kilograms (or Litres) per cubic metre. If ice is used for temperature control, it shall be expressed in incremental units (sacks, bags, or pails, etc.) and kilograms per cubic metre and shall be included in the maximum allowable water quantity.
- Admixtures dispensed as liquids shall be expressed as Litres per cubic metre and where applicable, expected dosage range stated.
- The quantity of any pre-measured, pre-packaged additives, such as fibres, shall be expressed in incremental units (sacks, bags, or pails, etc.) and kilograms per cubic metre.
- The absolute volume of each material, air content and the total sum of the absolute volumes of all materials shall be provided in cubic metres (m^3).
- Slump, unit weight, air content, strength characteristics, and any other specified characteristic shall be reported for each set of mixture proportions intended for use.

Incomplete submittals by the Designer may result in rejection of the mix design submittal.

211.03.05 Trial Mixes. The Contractor shall produce test batches of 4.0 m^3 volumes on all proposed bridge deck mix designs. The Contractor shall produce test batches of 4.0 m^3 volumes on other mix designs where deemed necessary by the Ministry Representative. Point of discharge sampling and testing shall be carried out by the Contractor to verify the pertinent parameters of the proposed mix design. If placement is by pump, concrete shall be sampled and fully tested at both the truck chute discharge and at the pump discharge locations. Sampling and testing shall be in accordance with DBSS 211.09, Table 211-J and Table 211-K. Test parameters include all required properties of the fresh and hardened concrete, workability and finishability of the mix, strength gain characteristics, including 7 and 28-day compressive strengths, and any other additional testing of the hardened concrete that may be required by the Ministry Representative. Batching, haul times and conveyance procedures used for the test batch must emulate the procedures expected during the actual concrete placement. The Contractor shall be fully responsible for all costs associated with the test batches. Test batch trials that do not meet the specified requirements shall be repeated.

211.03.06 Adjustments to Mix Designs. If, during the progress of the work, the mix design is found to be unsatisfactory, the Designer shall make the necessary adjustments. Adjustments shall be submitted to the Ministry Representative for review prior to subsequent concrete placement. Notwithstanding the Ministry Representative's review of the proposed mix design, it remains the Contractor's responsibility to ensure that the concrete meets all Contract and Design requirements.

211.04 Materials.

211.04.01 General. All material properties used in concrete production shall conform to CSA A23.1, unless otherwise specified in these specifications. Laboratory testing shall be carried out by a testing laboratory certified in accordance with the requirements of CSA Standard A283. Field test procedures shall be undertaken by personnel certified under an industry-recognized program.

211.04.02 Cementitious Material. The Contractor shall obtain from the supplier, a mill test report and a manufacturer's certificate of compliance representing each delivered shipment and type of cementitious material. .

211.04.02.01 Portland Cement. Unless otherwise specified, the Contractor shall use Type GU (General Use) Portland Cement in all concrete. Cement shall conform to the requirements of CSA Standard A3001, Cementitious Materials for use in Concrete.

The total alkali content [Na_2O equivalent, (Na_2O + (0.658 x K_2O))] of the Type GU Portland Cement shall not exceed 0.60% by mass.

211.04.02.02 Silica Fume. Dry condensed silica fume shall conform to CSA Standard A3001 Type SF, with a SiO_2 content of at least 85%, a maximum of 10% ignition loss, and maximum SO_3 content of 1.0%. Silica Fume application rates shall be 8% maximum by mass of Portland Cement.

211.04.02.03 Fly Ash. Fly ash shall conform to CSA Standard A3001, Type F or C1, with a maximum CaO content of 8%, a maximum SO_3 content of 5%, a maximum 8.0% ignition loss for Type F and a maximum 6% ignition loss for Type C1. Fly ash application rates shall be limited to a maximum of 25% by mass of Portland Cement in substructure concrete. Fly ash shall not be used in bridge deck concrete without the acceptance of the Designer and Ministry Representative. Upon such acceptance the maximum application rate shall be 15% by mass of Portland Cement.

211.04.02.04 Blended Hydraulic Cement. Blended Hydraulic Cement shall not be used without the acceptance of the Designer and Ministry Representative.

211.04.03 Water. Water to be used for mixing concrete or mortar and for curing shall conform to the requirements of CSA Standard A23.1 and shall be free from contamination by oil, acid, alkali, organic matter, sediment or other deleterious substances. The Contractor shall not use water from shallow, stagnant or marshy sources.

211.04.04 Admixtures. Admixtures shall not be incorporated into the mix design and/or added to the concrete without the acceptance of the Designer and Ministry Representative.

Unless otherwise specified, all concrete shall contain Air-Entraining Admixtures.

Air-Entraining Admixtures shall conform to ASTM C260.

Chemical admixtures shall conform to ASTM C494.

Type F High Range Water Reducer (superplasticizer) shall be used when concrete contains silica fume.

All accepted admixtures shall be compatible with all other admixtures and constituents used in the mix.

Type B, Retarding or Type D, Water-Reducing and Retarding (Hydration Stabilizing) Admixtures shall not be incorporated into the mix design and/or added to the concrete without the acceptance of the Designer and Ministry Representative. When accepted by the Designer and Ministry Representative, the guidelines given in DBSS 211.08.02 shall apply.

Calcium chloride or any admixture containing chloride ions shall not be used. If requested, a written statement from the manufacturer stating that the admixture contains no intentionally added calcium chloride shall be provided to the Quality Manager and Ministry Representative.

211.04.05 Aggregates. All aggregates are to be natural, processed or manufactured granular material composed of hard, sound and durable particles, free of adherent coatings, shale, clay, organic materials and other soft or disintegrated pieces.

211.04.05.01 Coarse Aggregate. All coarse aggregate materials shall conform to the following requirements:

- Sampling shall be according to CSA Test Method A23.2-1A.
- Gradation shall be determined in accordance with CSA Test Method A23.2-2A and 5A. (All gradations shall be based on a washed sieve analysis, see Table 211-D footnotes).
- Gradation of the coarse aggregate shall be within the limits shown in Table 211-B for the particular size range or nominal maximum size of coarse aggregate specified.

NOTE: The gradation of coarse aggregates used shall be such that the percentage passing any one sieve does not vary more than 5% from the initially approved coarse aggregates used in the mix design.

- The maximum Petrographic Number (PN) of coarse aggregate shall not exceed 125, and shall be determined in accordance with CSA A23.2 15A.
- The amount of deleterious substances in coarse aggregate shall not exceed the limits prescribed in Table 211-E.
- Other specified physical properties measuring aggregate quality referenced in Table 211-D shall not exceed the limits prescribed in Table 211-E.

Table 211-B GRADATION REQUIRMENTS FOR COURSE AGGREGATES

NMSA (1) mm	Product Size, mm	Total Passing Each Sieve, Percentage by Mass								
		56 mm	40 mm	28 mm	20 mm	14 mm	10 mm	5 mm	2.5 mm	1.25 mm
40	40-5 (2)	100	95-100	__	35-70	__	10-30	0-5	__	__
28	28-5 (2)	__	100	95-100	63-83	30-65	__	0-10	0-5	__
20	20-5	__	__	100	90-100	50-90	25-60	0-10	0-5	__
14	14-5	__	__	__	100	90-100	45-75	0-15	0-5	__
10	10-2.5	__	__	__	__	100	85-100	10-30	0-10	0-5

(1) NMSA – nominal maximum size of course aggregate. Definition: The standard Sieve size opening immediately smaller than the smallest through which all of the aggregate must pass.

(2) To prevent segregation, aggregates that make up the above gradings shall stockpiled and batched in two or more separate sizes as per CSA A23.1

211.04.05.02 Fine Aggregate. Fine aggregate shall conform to the following requirements:

- Sampling shall be according to CSA Test Method A23.2-1A.
- Grading shall be within the limits given in Table 211-C when tested in accordance with CSA Test Method A23.2-2A and 5A. (All gradations shall be based on a washed sieve analysis)
- Fineness modulus shall be between 2.3 and 3.1
- Fineness modulus tolerance shall be ±0.2 (except that the above sentence shall apply) from the fineness modulus of the initially approved fine aggregate used in the design mix.
- A maximum 45% shall be retained between any two consecutive sieves.
- The amount of deleterious substances in fine aggregate shall not exceed the limits prescribed in Table 211-E.
- Other specified physical properties measuring aggregate quality referenced in Table 211-D shall not exceed the limits prescribed in Table 211-E.

211.04.06 Steel Fibres. Steel fibres shall meet the requirements of ASTM Standard A820 Type 1 Cold Drawn High Tensile Deformed Steel Fibres. A mill certificate showing compliance with ASTM Standard A820 shall accompany each delivered lot. Steel Fibres shall be Xerox 1 or Novotex or an alternate acceptable to the Ministry Representative. Concrete where steel fibres are used shall contain 50 kg of 30 mm or 50 mm length fibres per cubic metre of concrete.

TABLE 211-C GRADATION REQUIREMENTS FOR FINE AGGREGATES

SIZE mm	TOTAL CUMULATIVE PASSING EACH SIEVE, PERCENTAGE BY MASS	SIEVE SIZE mm	TOTAL CUMULATIVE PASSING EACH SIEVE, PERCENTAGE BY MASS
10	100	0.630	25-65
5	95-100	0.315	10-35
2.5	80-100	0.160	2-10
1.25	50-90		

TABLE 211-D REQUIRED AGGREGATE TESTING FOR NORMAL DENSITY COARSE AND FINE AGGREGATE (PER INDIVIDUAL PRODUCT & AGGREGATE SOURCE)

Test Method	Test Description	Test Data Validity[5]
CSA A23.2-2A[1] CSA A23.2-5A[1]	Sieve Analysis of Fine and Coarse Aggregate Amount of Material finer than 80 µm in Aggregate	Within 90 days[6] Within 90 days[6]
CSA A23.2-3A	Clay Lumps in Natural Aggregate	Within 3 years
CSA A23.2-4A	Low Density Granular Material in Aggregate	Within 3 years
CSA A23.2-6A, 12A	Relative Density and Absorption of Fine and Coarse Aggregate	Within 1 year
CSA A23.2-7A	Test for Organic Impurities in Fine Aggregates for Concrete	Within 2 years
CSA A23.2-8A[3]	Measuring Mortar-Strength Properties of Fine Aggregate	Within 2 years
CSA A23.2-9A[2]	Soundness of Aggregate by Use of Magnesium Sulphate	Within 4 years
CSA A23.2-14A	Potential Expansivity of Aggregates (Procedure for Length Change Due to Alkali-Aggregate Reaction in Concrete Prisms)	Within 5 years
CSA A23.2-13A	Flat and Elongated Particles in Coarse Aggregate - Procedure A (Length to Width Ratio 4:1)	Within 3 years[6]
CSA A23.2-16A[7]	Resistance to Degradation of Small-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine	Within 4 years
CSA A23.2-23A[2]	Test Method for the Resistance of Fine Aggregate to Degradation by Abrasion in the Micro-Deval Apparatus	Within 4 years
CSA A23.2-29A[2]	Test Method for the Resistance of Coarse Aggregate to Degradation by Abrasion in the Micro-Deval Apparatus	Within 4 years
CSA A23.2-15A[4]	Petrographic Analysis of Coarse & Fine Aggregate specific for use as concrete aggregate; and Determination of the Petrographic Number	See Note 4

(1) Each aggregate product that may be used as a component of the proposed mix must have a washed sieved analysis performed showing the cumulative percent passing for each of the following sieve sizes; 56, 40, 28, 20, 14, 10, 5, 2.5, 1.25, 0.630, 0.315, 0.160 and 0.080 mm. Sieve analysis reports are to include all relevant sampling information (date/sampled by/location/field sample size) and include the actual test specimen sample mass.

(2) Test Method A23.2-9A can be performed as an alternative to performing Test Method A23.2-23A and A23.2-29A. Refer to CSA A23.1, Section 4.2, Table 12 for further information.

(3) Test Method A23.2-8A is required only if the aggregate fails to meet the requirements of Test Method A23.2-7A as described in CSA A23.1 Section 4.2.3.3.3.2

(4) Petrographic Assessment of Aggregates

The Contractor shall submit the results of the Petrographic Analysis, certified by a qualified professional registered with the Association of Professional Engineers and Geoscientists of B.C. and indicating the aggregate's suitability for its intended use.

The petrographic analysis shall not be used to forfeit the requirement of performing other necessary aggregate quality testing.

(5) Test data validity requirements are based on time periods prior to the expected initial date of concrete production. It is only applicable to aggregate sources proven to be materially consistent, based solely on historical test data. The Ministry Representative, based on submitted test data and/or visual observations, may reduce the above listed time periods.

(6) Quality control during aggregate production shall include the following test methods at the specified test frequencies: CSA A23.2-2A & A23.2-5A (1 per 500 tonnes), and A23.2-13A (1 per 1000 tonnes).

(7) Test Method CSA A23.2-16A can be waived if CSA A23.2-29A test results meet the specified criteria.

TABLE 211-E LIMITS FOR DELETERIOUS SUBSTANCES IN AND PHYSICAL PROPERTIES OF AGGREGATES

Test Method	Substance or Property	Maximum Percentage by Mass of Total Sample	
		Coarse Aggregate	Fine Aggregate
CSA A23.2-3A	Clay Lumps	0.25	1.0
CSA A23.2-4A	Low-Density Granular Materials	0.5	0.5
CSA A23.2-5A	Material Finer than 0.08 mm for Concrete Subject to Wear	1.0	3.0
	Material Finer than 0.08 mm for Other Concrete	1.0	5.0
CSA A23.2-9A	Magnesium Sulphate Soundness Loss	12	16
CSA A23.2-16A,	Los Angeles Abrasion Loss for Concrete Subject to Wear	35	N/A
CSA A23.2-23A-29A	Micro-Deval Abrasion Loss	17	20
CSA A23.2-12A	Absorption Percentage	2.0	N/A
CSA A23.2-7A	Organic Impurities in Fine Aggregate	N/A	Standard colour or lighter (1)
CSA A23.2-13A	Flat and Elongated Particles in Coarse Aggregate - Procedure A (Length to Width Ratio 4:1)	20	N/A
CSA A23.2-14A	Alkali-Reactivity Expansion in Concrete Prisms	0.040 (2)	0.040 (2)

(1) Standard Colour is defined as Organic Plate No. 3

(2) Expansion measured at 1 year

211.05 Storage and Handling of Materials - General. The storage and handling of all materials used in the production of concrete shall conform to CSA A23.1, unless otherwise specified herein. All materials shall be stored and handled in a manner that will prevent contamination or deterioration, otherwise they shall be rejected. Access shall be provided to the storage facilities to allow for inspection and sampling. Where applicable, materials shall be identified by MSD sheets.

211.05.01 Cementitious Material. All cementitious materials shall be stored in a suitable weather-tight structure capable of protecting the materials from dampness, hydration and contamination, otherwise they shall be rejected. Materials shall be free of lumps at all times during their use in the work and shall have a maximum temperature of 50°C.

211.05.02 Tanks. Tanks used for storing and/or hauling, concrete mixing and curing water shall be free

from contamination by oil, acid, alkali, organic matter, sediment, rust or other deleterious substances.

211.05.03 Admixtures. All admixtures shall be stored in manufacturers' original labelled containers with clearly legible labels and be kept above freezing at all times and in accordance with the manufacturers' technical data sheets. Admixtures damaged by any cause shall be rejected.

211.05.04 Other Materials. Other materials, such as steel fibres, etc. shall be stored and handled in accordance with the manufacturer's recommendations and instructions.

211.05.05 Aggregates. Each nominal size of aggregate, including coarse and fine shall be separately stockpiled, stored and handled in a manner that will prevent contamination, intermixing and segregation. A minimum of 10 m³ of each aggregate size, in excess of the requirements of the daily concrete operations, shall be maintained in identifiable stockpiles at the batch site in order to ensure proper continuity of the work with approved aggregates.

NOTE: Stockpiled aggregate, which has segregated, contaminated or intermixed with foreign matter of any kind, shall be rejected. Separate batch plant storage bins shall be provided for each size of aggregate to be batched.

Precautions shall be taken when moving, handling and transferring aggregates to prevent contamination, segregation or degradation in accordance with the following recommendations:

- Aggregate blanket 150 mm thick should be left on the ground as stockpiles are depleted.
- Aggregate transfer over stockpiles or storage bins of other sized aggregates should be avoided.
- Storage bins should be filled only to 150 mm from the top unless a suitable extension rim is provided to prevent contamination.
- Front-end loader buckets should be filled by raising them vertically up the face of the stockpile, not by thrusting into the stockpile in one place.
- Depositing aggregates into the batch plant storage bins should be made directly over the bins' outlet.
- Storage bins should be kept as full as possible during batching to minimize segregation as the aggregates are withdrawn from the bins.

Acceptable and uniform aggregate moisture content shall be obtained and maintained by necessary means to include the following:

- Washed or wetted aggregates should remain in stockpiles for a minimum of 12 hours to permit a uniform moisture content throughout the stockpile to be reached before aggregate transfer to batch plant storage bins.
- After each day's concrete production during wet weather, fine aggregate bins should be either emptied or covered with a suitable waterproof covering.

211.06 Batching

211.06.01 Batching Equipment

Batching equipment shall comply with the requirements of CSA A23.1.

Equipment specifications in DBSS 211.06.01 (i) to (vi) below are guidelines only, and as such, are not mandatory.

i) General - Batching equipment, such as weighing mechanisms, gates, water lines and dispensing systems, to be maintained in good working order. Batching bins to be completely emptied of all material before the first and succeeding batches are measured.

Batching equipment to have the following capabilities and facilities:

- bins or silos for storage of aggregates, silos for storage of cement and pozzolan
- free movement of each type or size of material to discharge openings without contamination
- design of storage and hauling facilities to prevent:
  - loss or intermingling of different sizes and types of materials
  - contamination by deleterious substances
  - harmful segregation and breakage
- covers to stockpiles of fine aggregates to ensure constant moisture content
- all fulcrums, clevises and similar working parts maintained in clean condition
- provision made, in remote control plants and/or where batch cycles are timed, for the complete filling and discharge of the measuring unit for each batch.

ii) Batch Charging Mechanism - Batch Charging Mechanism requirements to be as follows:

- material flow control within the specified tolerances
- hopper construction to eliminate material accumulation and to permit the complete discharge of every batch
- cement weighing hopper:

- self-cleaning and properly ventilated to allow air to

escape

- accessible for inspection
- dust sealed between the charging mechanism and the hopper to maintain weighing accuracy.

iii) Scales and Dispensing Mechanism - Scales and Dispensing Mechanisms to conform to the following requirements:

- beam type or springless dial type
- other methods of weighing (electric, hydraulic, load cells, etc.) are acceptable provided the specified weighing tolerances and accuracy requirements are met
- all accurate to ±0.4% of the total capacity of the scale when static load tested
- zero balance adjustment capability
- unaffected by binding or vibration due to vibrators or other appurtenances
- working range between 10% and 90% of the scale reading
- equipped with prominent markers with individual batch weight setting capabilities for dial scales
- calibration facilities including an adequate number of standard test masses
- beam type scales with provision for indicating to the operator that the required load in the hopper is being approached; the device shall indicate at least the last 100 kg of the load.
- all weighing and indicating devices shall be in full view of the operator while charging the hopper, and the operator shall have convenient access to all controls.

iv) Volumetric Devices for Water Measurement - Volumetric devices for water measurement to be:

- fitted with such valves and connections as are necessary to divert the water measured for a batch to easily verify the accuracy of measurement
- arranged so that measurements will be unaffected by variable pressures in the water supply line
- calibrated to an accuracy of 2% of the batch volume.

v) Dispensing Systems for Liquid Admixtures - Dispensing systems for liquid admixtures, other than weigh systems, to have the following capabilities:

- visual, volumetric measuring or readout units located as close as possible to eye level for easy reading by the operator
- positioned in such a manner that discharge into the batch is observed by the operator from a normal working position
- volumetric measuring unit for periodic check of dispenser accuracy where a positive displacement system is used
- visual measuring unit for an accurate visual check of increments of 30 mL of air entraining agent or of chemical required to treat a maximum of 25 kg of cement

vi) Certificate or Report of Inspection - A certificate or report of inspection, from a Weights and Measures approved, independent testing authority, and which is not more than one year old for the plant in its present position, shall be on display in the plant at all times. The certificate or report will be accepted as proof of accuracy of the scales or weighing devices. Where there is reasonable doubt concerning the accuracy of the scales or weighing devices the Quality Manager or Ministry Representative may require their recalibration and recertification, at the Contractor's expense, during progress of the work. Plant relocation or major alterations shall require such recalibration and recertification.

211.06.02 Batching of Materials. Material may be weigh-batched separately or cumulatively.

Aggregates shall be batched by mass, cement and mineral admixtures in the powder form by mass or bag.

The mass of any ingredient to be batched shall not be less than 10% nor greater than 90% of the scale capacity.

When batched by 40 kg bags, only full and sealed bags of cement shall be used with no fractional bags permitted. Cement batched by mass shall be to an accuracy of ±1% for batch quantities between 30% and 90% of the scale capacity. When the mass of cement being batched is between 10% and 30% of the scale capacity, the mass of cement batched shall not be less than the required mass nor more than 4% in excess. Intermediate bulk cement shall be weighed to an accuracy of ±1%.

When aggregates are measured by mass, batch masses shall be based on the required mass of saturated surface dry aggregate corrected for the moisture conditions of the aggregate at the time of batching. Field determination of free moisture in the aggregates shall be made by any proven method such as CSA A23.2-11A or by the use of a device such as the "Speedy Moisture" gauge. Where doubt exists as to the accuracy of such methods, total moisture in the aggregate shall be determined in accordance with ASTM C566. The minimum test frequency shall be once per each day's production. When individual aggregate weigh batchers are used, the scale reading for each material shall be within 2% of the specified mass. In a cumulative aggregate weigh batcher,

the cumulative mass after each measurement shall be within 1% of the required cumulative amount when the scale is used in excess of 30% of its capacity. For cumulative measurement less than 30% of scale capacity, the allowable variation shall be ±0.3% of scale capacity or ±3.0% of the required cumulative mass, whichever is less.

Mixing water shall be measured by mass or volume to an accuracy of ±1%. Adjustments for free water contained in batched materials, such as free water in the aggregates, water contained in admixture solutions, shall be made to the quantity of water to be batched. The total amount of mixing water in the batch obtained from all sources shall be within ±3% of the specified quantity and shall not vary the designed water-to-cementitious material ratio of the mix by more than ±0.02. Mixers shall be completely emptied of all water prior to the loading of a concrete batch.

Admixtures shall be batched in liquid form by either mass or volume; in powdered form by mass or bag. Volumetric measurements of admixtures or air entraining agents shall be to an accuracy of ±3% of the required amount or 30 mL, whichever is greater. Measurement by mass shall be to an accuracy of ±3% of the required amount.

The addition of cement to a fully-batched load of aggregates will not be accepted.

211.07 Mixing.

211.07.01 General - Sufficient plant capacity and transporting equipment shall be provided by the Contractor to ensure continuous delivery of concrete at the rate required with the necessary intervals between batches, for the proper placing and finishing of the concrete without the formation of cold joints in the finished concrete.

211.07.02 Concrete Mixers.

i) General - Concrete mixers shall conform to the following requirements:

- manufacturer's rating plate carried in a prominent position, indicating rated mixing capacity, and recommended speed of mixing which must be in the range of not less than 4 nor more than 18 rpm
- demonstrated capability of operating satisfactorily at the recommended mixing speed
- number of revolutions of the drum or blades registered with a counter in working condition
- charge and discharge openings and chute free from appreciable accumulations of cement or concrete, and hopper and chute surfaces clean and smooth
- batch water measuring equipment in good operating condition.

ii) Volumetric Mobile Mixers. - Volumetric mobile mixers shall conform to the following requirements:

- Shall be auger type mixers
- The mobile mixers shall be calibrated with the actual cement and aggregate products used in the concrete mix. Calibration tables and curves shall be submitted to the Quality Manager and Ministry Representative. The volume of the concrete produced shall be confirmed and shall be within 2%.
- Mobile mixers shall have separate flow controlled supply hoppers for each individual aggregate gradation and cementitious constituent material to be added volumetrically.
- May be used provided that the accuracy of batching and uniformity of concrete is as required by CSA A23.1.
- Cement flow rates shall be checked before (during if required) each placing operation.
- The scale being used to calibrate the mobiles shall have a minimum capacity of 50 kg.
- The water tank shall have a stand pipe showing the amount of water in the tank.
- The mobiles shall have individual flow meters and admixture tanks for each admixture to be used as part of an admixture injection system.
- The mobiles shall be able to advance backwards and forwards from controls located at the back of the mobile.
- The mixer auger shall be maintained at minimum of a 25° angle from horizontal when mixing and discharging, unless otherwise indicated by the manufacturer.

211.07.03 Mixing Concrete. Mixing drums shall be clean and empty before being charged. The drum shall be rotated at the manufacturers' recommended mixing speed during charging and mixing. Concrete shall be mixed to the uniformity requirements of CSA A23.1, Section 5.2.3.5. When a satisfactory mixing time is established it shall be maintained for all batches of the same design, mixed with the same equipment.

211. 08 Delivery.

211.08.01 General - Off-site mixed concrete shall, after being mixed, be transported to the Site in either truck agitators or truck mixers operating at the speed designated by the manufacturer of the equipment as agitating speed.

When concrete is being mixed in mixer trucks while in transit, the speed of the mixing drum shall be reduced to agitating speed once the mixing time has elapsed. Delivery of mixed concrete in non-agitating equipment

will not be permitted.

211.08. 02 Time of Placement

Time of placement guidelines for bridge decks including diaphragms, bridge parapets, bridge medians, bridge sidewalks and approach slabs are given in DBSS 413, Bridge Decks and Concrete Overlays.

All other concrete shall be fully discharged and placed within 90 minutes after water and cement have been combined. Under conditions contributing to rapid stiffening of concrete the Contractor shall adjust its placing operations to suit the reduced handling time.

Time of placement extensions will be considered on a case by case basis and may require the use of Hydration Stabilizing Admixtures (HSA's) conforming to the requirements of ASTM C494 Type B, Retarding or Type D, Water-Reducing and Retarding Admixtures. If accepted by the Designer and the Ministry Representative, guidelines for the use of HSA's shall be as follows:

- Concrete shall be fully discharged and placed within 3 hours after water and cement have been combined.

- When HSA's are used, these time extensions are subject to preconstruction trials being conducted by the Contractor to establish the appropriate HSA dosage to provide suitable extended slump life of concrete without increasing the water/cementitious ratio of the concrete above that which would be required if HSA's were not used. The use of HSA's shall in no instance modify the maximum concrete temperature required at time of placement. A one time only addition of HSA will be allowed, this will be during initial batching of the concrete or immediately at completion of batching as recommended by the admixture manufacturer. Addition of HSA at any other time will be cause for rejection of the concrete.

211.08.03 Temperature Control - Temperature control guidelines for bridge decks including diaphragms, bridge parapets, bridge medians, bridge sidewalks and approach slabs are given in DBSS 413, Bridge Decks and Concrete Overlays.

The temperature of all other concrete, from the time of batching to complete discharge shall remain between 10° C and 25° C. Any concrete containing silica fume shall not have a maximum temperature in excess of 17°C at time of placement. Concrete used in elements having a minimum section thickness greater than 1000 mm shall have temperature controls as per CSA A23.1 Clause 5.2.4.4. Methods used for maintaining temperature controls shall not diminish the quality of the concrete, and shall not alter nor exceed the specified maximum W/C_m ratio.

When the atmospheric temperature is 20°C or higher, DBSS 211.18 – Hot Weather Concreting shall apply.

211.08.04 Addition of Water - No water shall be added after the initial introduction of the mixing water for the mix, except at the start of discharge within the initial 10% of the total load volume, and when:

- the specified water-to-cementitious materials ratio is not exceeded
- the measured slump is less than that specified
- no more than 60 minutes have elapsed after water and cement have been combined
- the concrete is in a mixer or truck mixer.

An amount not exceeding 16 L of water per cubic metre of concrete or 10% of the mix design water whichever is less may be added. After addition of water, the drum or blades shall be turned an additional 30 revolutions or more if necessary at mixing speed. The amount of water added shall be recorded on the delivery ticket. Water shall not be added to the batch at any later time. The Contractor assumes full responsibility for the on-site addition of water and its subsequent effect on the quality of the concrete.

211.08.05 Addition of Superplasticizer (HRWR) – When superplasticized concrete falls below the designated slump due to delay, it shall be re-tempered with superplasticizing admixtures only, not water, and shall only receive a maximum of one re-tempering. The amount of superplasticizer added shall be recorded on the delivery ticket. The Contractor assumes full responsibility for the on-site addition of superplasticizer and its subsequent effect on the quality of the concrete. The use of superplasticizers shall not be used for extending the time of placement requirements described in DBSS 211.08.02.

211.08.06 Control of Air Content on the Job Site – The air content of the concrete shall, if necessary, be brought up to the specified range by the addition of an air-entraining agent in the field. Mixing shall follow to ensure proper dispersion and upon completion of mixing, the air content shall be retested. The amount of admixture added shall be recorded on the delivery ticket. The Contractor assumes full responsibility for the on-site addition of the air entraining agent and its subsequent effect on the quality of the concrete.

211.08.07 Delivery Ticket - Concrete delivered to the job site shall be accompanied by a ticket providing the Contractor, Quality Manager and Ministry Representative with legible information, as follows:

- serial number of ticket
- date, name and location of concrete supplier
- name of the Contractor

- specific job designation (name and location)
- specific designation of concrete mix design
- truck number and volume of concrete in cubic metres
- addition of any materials added at the job site, at the request of the Contractor including but not limited to water and all chemical admixtures.
- time concrete was batched, arrived at job site, and completely discharged.
- amount of water added during batching including free moisture in aggregate, the maximum allowable amount of water that can be added at the site to maintain the specified maximum W/C_m ratio.

211.09 Quality Control and Quality Assurance Inspections, Testing and Acceptance.

211.09.01 General. The Contractor shall be fully responsible for hiring, scheduling, overseeing, performing and documenting all quality control and quality assurance testing and inspection in full compliance with the Contract.

The Ministry may perform on-site sampling and testing as a function of the Ministry's quality audit. Any Ministry quality testing shall not relieve the Contractor of responsibility for providing quality control and quality assurance.

211.09.02 Sampling and Testing. Laboratory testing shall be carried out by a testing laboratory certified in accordance with the requirements of CSA Standard A283. Field test procedures shall be undertaken by personnel certified under an industry-recognized program. Field inspectors shall have extensive experience in on-site quality control testing of concrete, with the applicable admixtures being used, and with on-site batch adjustments. Quality control testing personnel shall be on-site and available to test concrete for the complete duration of any concrete placement operation. Sampling and testing shall be in accordance with the test methods and minimum frequency levels as listed in Table 211-J and Table 211-K. Any test outside the specified range shall be immediately reported to the Contractor, Quality Manager and Ministry Representative. Copies of all test results shall be received by the Quality Manager and Ministry Representative within 1 day of the test date.

211.09.03 Sampling - Point of Acceptance. Determination of concrete properties for acceptance will be made based on samples taken at the final discharge end of the placement system. The Quality Manager and Ministry Representative may accept sampling at the truck discharge end provided that testing during the test batches confirm that the concrete mix parameters are within the stated specification requirements at the actual placement point. The Quality Manager or Ministry Representative may require additional testing to be performed at the point of placement at a predetermined frequency throughout the placement period. Such additional testing, including all costs, shall be deemed as part of the Contractor's quality control program.

211.09.04 Strength Tests. Unless otherwise specified in the Contract, a strength test shall be the average 28-day strength of three standard 100 mm x 200 mm test cylinders, sampled, cast, cured, transported, and tested in accordance with CSA A23.2- 3C and 9C. The cylinder size shall however, meet the aggregate size limitations specified in Clause 6.2 of CSA Test Method A23.2-3C. For each strength test, a fourth cylinder shall be cast for a 7-day strength determination. If one specimen in a strength test, in the opinion of the testing agency shows evidence of improper sampling, casting, or testing, including damage from improper handling and transporting (and if agreed to by the Ministry Representative), it shall be discarded and the remaining two specimen strengths averaged. Additional cylinders may be cast, at the discretion of the Ministry Representative or Quality Manager.

The strength of each classification of concrete shall be considered satisfactory if:

- the averages of all sets of three consecutive strength tests equal or exceed the specified strength
- no individual test shows strengths less than 90% of the specified strength

Each "classification of concrete" will be specified in the Design. Where fewer than three strength tests are taken, the results of one test or the average of two will be used.

TABLE 211-J STANDARD TEST METHODS FOR SAMPLING AND TESTING

Test Methodology	Test Description	Minimum Test Frequency
CSA A23.2-1C	Sampling Plastic Concrete	As stated below
CSA A23.2-3C CSA A23.2-9C	Making and Curing Concrete Compression and Flexural Test Specimens Compressive Strength of Cylindrical Concrete Specimens	One set for every 35 m^3 or portion thereafter for each classification of concrete placed within an individual structural element or component, and placed on any one-calendar day from a single supplier. The Quality Manager or Ministry Representative may request additional sets.
CSA A23.2-4C CSA A23.2-5C ASTM C1064	Air Content of Plastic Concrete by the Pressure Method Slump and slump flow of Concrete Temperature of Fresh Concrete	Every individual load unless approved otherwise by the Quality Manager and Ministry Representative Retests shall be performed following any load adjustments.
CSA A23.2-6C	Density, Yield, and Cementing Materials Factor of Plastic Concrete	Density and Yield tests shall be performed with every air test and strength test.

TABLE 211-K ADDITIONAL TEST METHODS FOR CONCRETE CONTAINING HIGH RANGE WATER REDUCERS (SUPERPLASTICIZERS)

Test Methodology	Test Description / Sample Requirements	Minimum Test Frequency
ASTM C457 *	Microscopic Determination of Parameters of the Air-Void System in Hardened Concrete	One test per individual test batch for each classification of concrete. Acceptance criteria shall be in accordance with CSAA23.1 Section 4.3.3.2 and 4.3.3.3. Further testing, during scheduled concrete pours may be required, as directed by the Quality Manager and Ministry Representative
ASTM C1202	Electrical Indication of Concrete's Ability to Resist Chloride Ion Penetration (Test age at 56 days)	One test per individual test batch for each classification of concrete Acceptance criteria shall be a coulomb rating not exceeding 1500 coulombs Each test set includes two 100 mm x 200 mm [4" x 8"] cylinders, sampled at the final placement discharge point. Casting, transporting and curing shall be in accordance with DBSS 211.09.05.

* for superplasticized concrete only

211.09.05 Test Cylinders. Making and curing concrete test cylinders shall be carried out in accordance with CSA A23.2-3C, except that the time for cylinders to reach the testing laboratory shall be between 20 and 48 hours from initial time of casting. The test cylinders shall be cast by the Contractor using standard CSA approved moulds. The

Contractor shall provide properly designed temperature-controlled storage boxes for test cylinder storage, as specified in CSA A23.2-3C Clause 7.3.2.1, for a period of at least 20 hours, and further protection from adverse weather and mishandling before and during delivery to the testing laboratory for curing and testing. The Contractor shall provide a maximum-minimum thermometer for each storage box.

The Contractor shall deliver the test cylinders to a CSA certified testing laboratory. Handling and transporting of the cylinders shall be in accordance with CSA A23.2-3C. No extra laboratory curing time will be allowed for cylinders that are delivered late to the laboratory. A copy of the test results shall be forwarded to the Quality Manager and Ministry Representative within 2 days of each individual strength test date. Test results shall include all information as described in CSA A23.2, Annex B, Form for Reporting Compressive Strength of Concrete Test Cylinders.

211.09.06 Field Cured Test Cylinders. The Quality Manager or Ministry Representative may require field cured test cylinders be cast to check the adequacy of the Contractor's curing or cold weather protection. The Contractor may also cast field cured test cylinders for determining interim strengths that may be required for further work progress approval such as opening to traffic, formwork removal and/or erection, etc.

Casting, handling, transporting and testing shall be in accordance with CSA A23.2-3C and 9C.

Test cylinders shall be stored as near as possible to the point in the structure that the test cylinders represent, and shall be afforded the same temperature protection and moisture environment as the structure.

At the end of the curing period the test cylinders shall be left in place, exposed to the weather in the same manner as the structure.

All test cylinders shall be removed from the field storage and stored in lime water at 23 ± 2ºC for 24 ± 4h immediately before time of testing to ensure uniform moisture conditions from cylinder to cylinder.

Test results on field cured test cylinders shall not be used as a basis for acceptance or rejection of the concrete.

211.09.07 Failure to Meet Minimum Strength Specifications.

i) Understrength Concrete - where the strength of concrete is not satisfactory as specified in DBSS 211.09.04, the Designer shall prepare a plan for remedial measures and the plan shall be submitted to the Ministry Representative for review. If the proposed plan is not acceptable to the Ministry Representative then the Ministry Representative may order replacement or reinforcing or other remedial measures , at the Contractor's expense, of portions of the structure. Alternatively, at the Ministry Representative's discretion, the understrength concrete may be accepted at a reduced price.

ii) Coring - the coring of concrete, to verify its strength, will be allowed only for concrete, which is otherwise to be replaced or reinforced. Coring will not be allowed for concrete for which reduced payment is to be made for understrength. When coring is allowed, cores shall be taken at locations directed by the Ministry Representative, who shall be present during the coring.

Where coring is allowed, it shall be carried out by an independent qualified testing firm. Three cores shall be obtained and tested in accordance with CSA A23.2-14C "Obtaining and Testing Drilled Cores for Compressive Strength Testing" at each test location. Concrete in the area represented by the core tests may be considered structurally adequate if:

- the average strength test result of each set of three cores from the portion of the structure in question is equal to at least 100% of the specified strength
- the strength test result of any single core is not less than 80% of the specified strength

Notwithstanding the final result, the Contractor will be responsible for all costs associated with the coring operation and testing, and shall not be reimbursed for inconvenience or other associated costs.

211.09.08 Failure to Meet Slump or Air Content Specifications. In the event that the slump and/or air content are outside the specified limits, the Ministry Representative may accept adjustments to correct the deficient condition as an alternative to rejection. In such cases, adjustments must be completed within the maximum time allowed as specified in DBSS 211.08.02 and additional testing shall be required to verify specification compliance. If compliance is not achieved, the concrete shall be rejected.

211.10 Placing of Concrete.

211.10.01 General - Equipment for conveying concrete at the job site, such as buggies, buckets, hoppers, chutes, belts and pumps, shall be of such design, size and condition to deposit a continuous and adequate supply of concrete of the specified mix and consistency without segregation at the required locations.

The equipment specifications in DBSS 211.10.02 to DBSS 211.10.06 inclusive are guidelines only, and as such, are not mandatory. DBSS 211.10.07 is mandatory. Placing of concrete shall also comply with the requirements of CSA A23.1